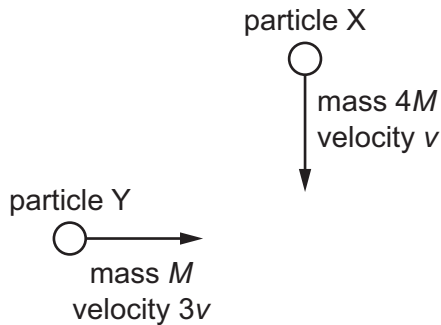


11 The diagram shows two particles in motion.

Particle X has mass $4M$ and is travelling with velocity v .

Particle Y has mass M and is travelling with velocity $3v$.

The paths of the particles are perpendicular.



The two particles collide and stick together to form a combined particle. No external forces act on the particles.

What is the magnitude of the momentum of the combined particle after the collision?

A Mv

B $4Mv$

C $5Mv$

D $7Mv$